

Unmatched precision magnetometer with ultra-low power, compact size, and an extensive dynamic range.

Datasheet - In Production



Description

The Siderion **BMM350** is a very small, high-performance, and low-current 16-bit 3-axis magnetometer.

The digital sensor is ideally suited for a wide range of consumer applications such as AR/VR, head orientation, indoor navigation, magnetic compass, high-end gaming, and dead reckoning.

The sensor is based on Bosch Sensortec's proprietary TMR technology and offers outstanding design flexibility, providing a single monolithic package solution.

Its JST-GH connectors make it ideal for integration with Pixhawk or other autopilot systems.

The BMM350 operates across a temperature range from -40°C to +70°C.

Features

- 5V - 3.5mA power supply
- High-resolution (~0.1μT)
- Wide Magnetic field range
- Compact footprint
- I²C interface

Applications

- Drones
- Ground vehicles
- Electronic compasses
- Steering angle sensors

Table 1. Device summary

Order ref code	Temperature [°C]	Product size [mm]	Weight [g]
SID-SENS-BMM350	-40 to +70	26.0 x 17.0 x 8.0	4

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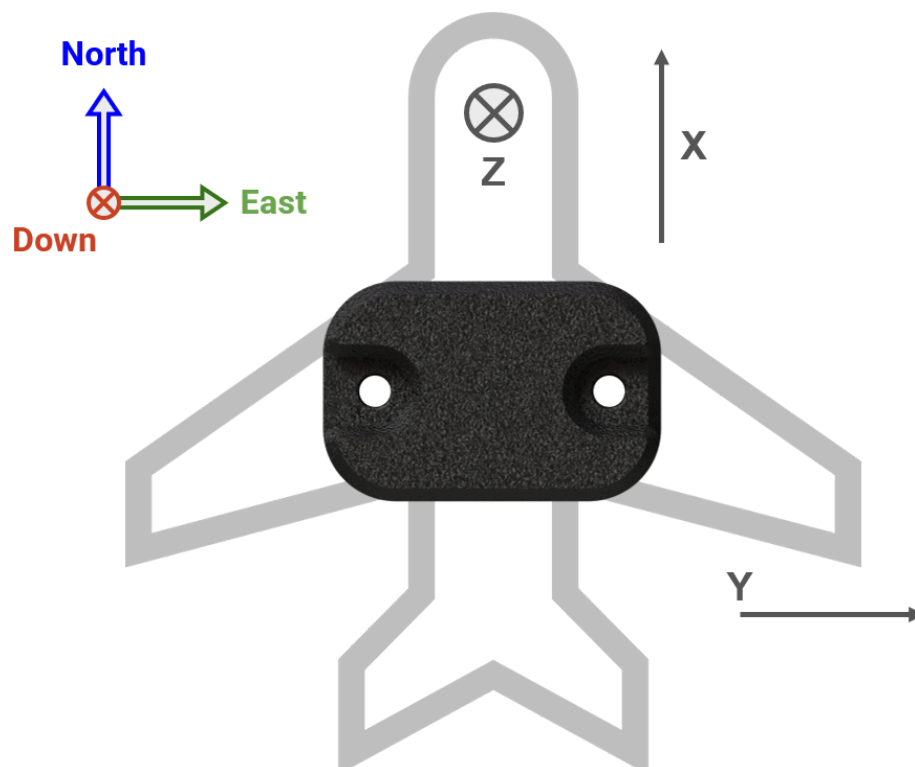
1. Block diagram and pin description

1.1 BMM350 Orientation

Figure 1. BMM350 - NED (XYZ) orientation (bottom view)



Figure 2. BMM350 - NED (XYZ) orientation (Top view on a drone)



1.2 Pin description

Figure 3. BMM350 pinout (side views)

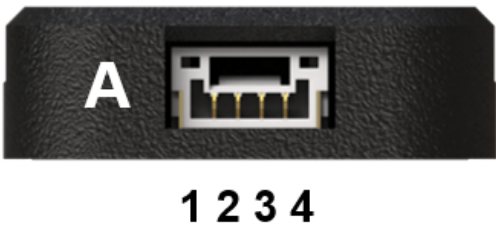


Table 2. Pinout configuration

	Pin	Name	Type	Function
I2C	B1	5V IN	P	5V input
	B2	I2C SCL	I	I2C Serial Clock
	B3	I2C SDA	I/O	I2C Serial Data
	B4	GND	P	Ground reference

P : Power / I : Input / O : Output

2. Absolute maximum ratings

Stresses above those listed as “absolute maximum ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 3. BMM350 absolute maximum ratings

Symbol	Parameter	Maximum value	Units
5V IN	5V input	-0.5 to +5.5	V
Vdd	Internal supply voltage	-0.5 to 3.6	V
Vdd_IO	I/O pins supply voltage	-0.5 to Vdd+0.3	V
Icc	Input current	3.5	mA
Mf	Magnetic field (any direction)	-2000 to +2000	μT
W	Product weight	4	g
TOP	Operating temperature	-40 to +70	°C
TSTG	Storage temperature	-40 to +80	°C



This device is sensitive to electrostatic discharge (ESD), improper handling can cause permanent damage to the part.

3. Specifications

3.1 Bosch Sensortech's BMM350 specifications

Table 4. BMM350 electrical specifications

Symbol	Parameter	Test conditions	Min.	Typ.	Max.	Units
Vdd	Internal supply voltage		1.72		1.98	V
Vdd_IO	Supply voltage for I/O		1.72		3.6	V
Idd	Current consumption				3.5	mA
Vil	IO pin low level input voltage		0		0.3* Vdd	V
Vih	IO pin high level input voltage		0.7* Vdd		Vdd	V
Iidle	Idle Mode Current				1.8	μA
Ileak	Leakage Current @ DVDD pin (AVDD=AVSS=DVSS=0V , DVDD=3.6V)				100	nA
Top	Operating temperature		-40		+85	°C
I2Cadd	7bit I2C address	0b0010100 (0x14)				

Table 5. BMM350 Geomagnetic Sensor Performance

Parameter	Min	Typ	Max	Units
Field Measurement Range	-2000		+2000	μT
Gain		17		LSB/ μT
Sensitivity		100		nT
Noise (X & Y axis)		190		nTrms
Noise (Z axis)		450		nTrms
Noise Density @400Hz (X & Y axis)		13.4		nT/ $\sqrt{\text{Hz}}$
Noise Density @400Hz (Z axis)		31.8		nT/ $\sqrt{\text{Hz}}$
Zero-B Offset		± 2		μT
ODR in normal mode	Typ -2%	400 200 100 50 25 25/2 25/4 25/8 25/16	Typ +2%	Hz
Maximum forced mode trigger frequency		200		Hz

4. Applications

4.1 Standalone configuration

5. Communication interfaces

The sensor offers an I2C communication interface. All the inputs have internal pull-up resistors in normal operation and can be left open if not used.

All the PIOs are supplied by Vdd.

5.1 Slave I2C interface

An I2C compliant interface is available for communication with an external host CPU. The interface can be operated in slave mode only. It is fully compatible with Fast-Mode of the I2C industry standard. Since the maximum SCL clock frequency is 400 KHz (in Fast-Mode), the maximum bit rate is 400 Kbit/s. The interface stretches the clock when slowed down while serving interrupts, therefore the real bit rates may be slightly lower.

The SDA and SCL lines do not have any internal pull-up resistors soldered in place. External pull-up resistors may be required for correct operation (typically 1 kΩ to 4.7 kΩ)

Figure 4. I2C chronogram specifications

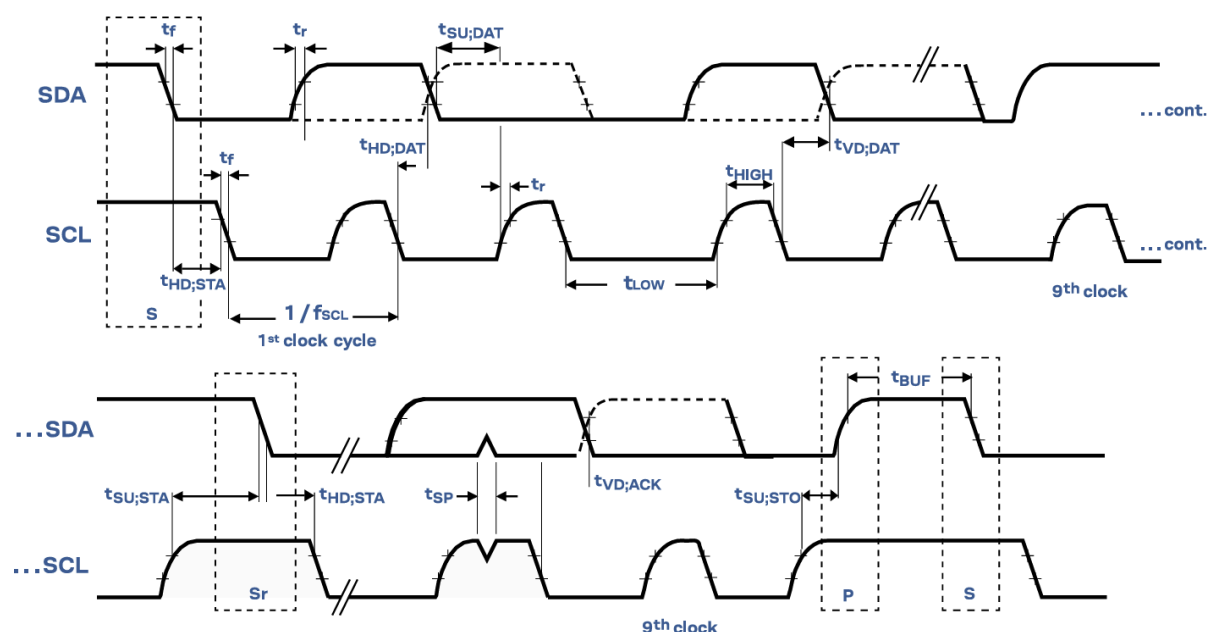
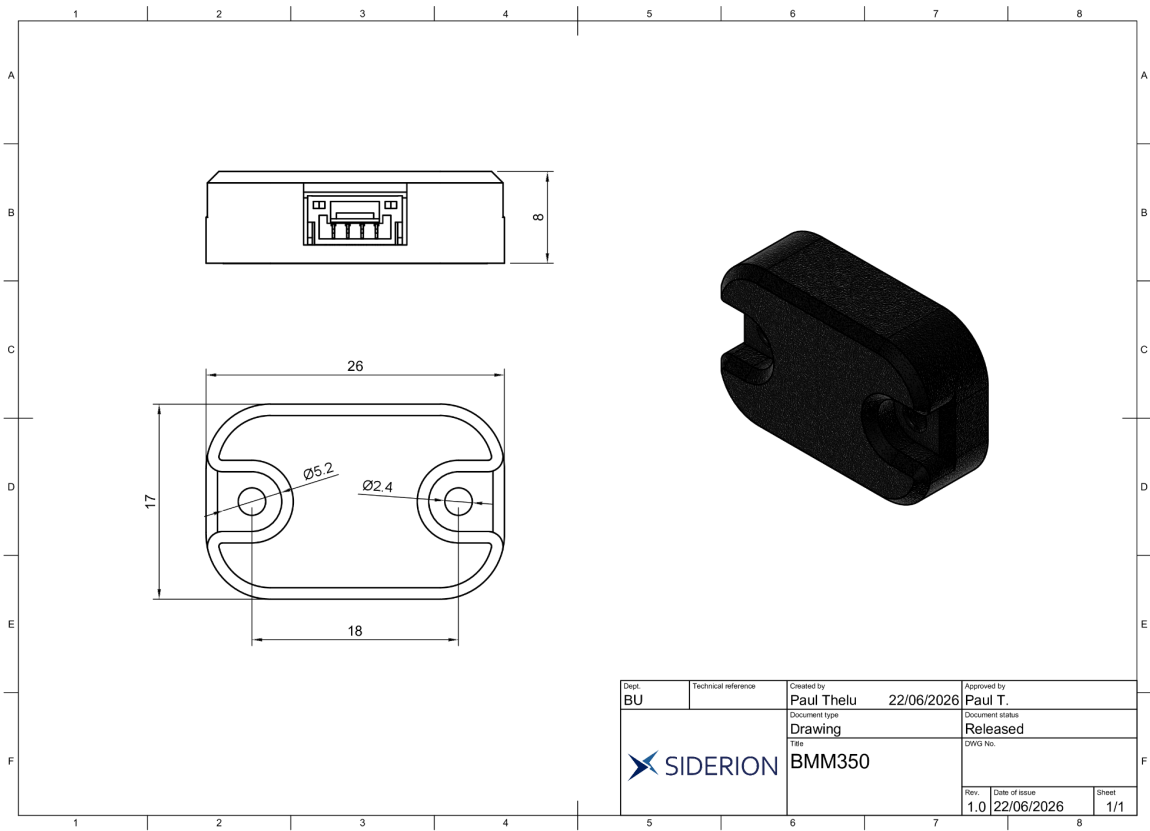


Table 5. BMM350 sensor I2C timing specifications

Symbol	Parameter	Min	Max	Units
Vil	LOW-LEVEL input voltage	0	0.2xVdd	V
Vih	HIGH-LEVEL input voltage	0.8xVdd	Vdd	V
Cb	Capacitive load for each bus line		20	pF
Fscl	SCL clock frequency	0	400	KHz

6. Mechanical drawings

Figure 6. BMM350 v1.0 mechanical drawings



7. Revision history

Table 8. Document revision history

Date	Revision	Changes
03-Aug-2025	1.0	SiderionDoc_BMM350 / Initial release

8. Appendix

Bosch Sensortech's datasheet: [BMM350 Datasheet](#)

Siderion user's guide:

Siderion tutorials:

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